

Chapter 5 Differential Entropy and the Gaussian Channel

Hui-Ming Wang

1

Structure



XI'AN JIAOTONG
UNIVERSITY

Chapter 0
Preview

Chapter 1
Entropy and Mutual
Information

Chapter 2
AEP and
Entropy Rate

Probability
theory
and
Stochastic
process

Chapter 3
Data Compression
(Source Coding)

Chapter 4
Channel Capacity and
Channel Coding Theorem

Discrete
R. V

Chapter 5
Differential Entropy and the
Gaussian Channel

Continuous
R. V

IIE,Xi'an Jiaotong University

2

2



Introduction

- So far, we have investigated various aspects of information theory, but we only considered **discrete random variables**.
- How to deal with the **continuous random variables**??
- In this chapter, we introduce the concept of **differential entropy** (微分熵), which is the entropy of a continuous random variable.
- We will also derive Differential entropy is similar in many ways to the entropy of a discrete random variable. But there are some important differences, and there is need for some care in using the concept. (跟熵有很多相似之处, 但是也有重要区别)
- e the famous formula operational capacity of additional white Gaussian noise (AWGN) channel.



Outline

- Differential Entropy
- Mutual Information
- Properties and Relations
- AEP
- Gaussian Channel
- Channel Capacity of Gaussian Channel
- Some Other Channels



Differential Entropy

Definition Let X be a random variable with cumulative distribution function $F(x) = \Pr(X \leq x)$. If $F(x)$ is continuous, the random variable is said to be continuous. Let $f(x) = F'(x)$ when the derivative is defined. If $\int_{-\infty}^{\infty} f(x) = 1$, $f(x)$ is called the *probability density function* for X . The set where $f(x) > 0$ is called the *support set* of X .

Definition The *differential entropy* $h(X)$ of a continuous random variable X with density $f(x)$ is defined as

$$h(X) = - \int_S f(x) \log f(x) dx,$$

where S is the support set of the random variable.

For discrete random variable, entropy *does not depend on the actual values, but only on the probabilities*. (熵的大小跟离散随机变量的取值无关, 而微分熵会跟连续随机变量的取值有关)

For continuous case, differential entropy *does*. ([1,2,3,4], [2,4,6,8])



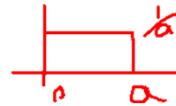
Differential Entropy

□ Differences with entropy

- Some times the density function does not exist for a random variables or the above integral does not exist.
- Differential entropy can be negative. (微分熵可能为负)

Example 8.1.1 (*Uniform distribution*) Consider a random variable distributed uniformly from 0 to a so that its density is $1/a$ from 0 to a and 0 elsewhere. Then its differential entropy is

$$h(X) = - \int_0^a \frac{1}{a} \log \frac{1}{a} dx = \log a.$$



Note: For $a < 1$, $\log a < 0$, and the differential entropy is negative. Hence, unlike discrete entropy, differential entropy can be negative. However, $2^{h(X)} = 2^{\log a} = a$ is the volume of the support set, which is always non-negative, as we expect.



Differential Entropy

□ Normal distribution 正态分布

Example 8.1.2 (Normal distribution) Let $X \sim \phi(x) = \frac{1}{\sqrt{2\pi}\sigma^2} e^{-\frac{x^2}{2\sigma^2}}$.
Then calculating the differential entropy in nats, we obtain

$$\begin{aligned} h(\phi) &= - \int \phi \ln \phi \\ &= - \int \phi(x) \left[-\frac{x^2}{2\sigma^2} - \ln \sqrt{2\pi}\sigma^2 \right] \\ &= \frac{EX^2}{2\sigma^2} + \frac{1}{2} \ln 2\pi\sigma^2 \\ &= \frac{1}{2} + \frac{1}{2} \ln 2\pi\sigma^2 \\ &= \frac{1}{2} \ln e + \frac{1}{2} \ln 2\pi\sigma^2 \\ &= \frac{1}{2} \ln 2\pi e\sigma^2 \quad \boxed{\text{nats.}} \end{aligned}$$



Differential Entropy

□ Joint Differential Entropy

Definition The differential entropy of a set $X_1, X_2, \dots, X_n$ of random variables with density $f(x_1, x_2, \dots, x_n)$ is defined as

$$h(X_1, X_2, \dots, X_n) = - \int f(x^n) \log f(x^n) dx^n.$$

Theorem 8.4.1 (Entropy of a multivariate normal distribution) Let $X_1, X_2, \dots, X_n$ have a multivariate normal distribution with mean μ and covariance matrix K . Then

$$h(X_1, X_2, \dots, X_n) = h(\mathcal{N}_n(\mu, K)) = \frac{1}{2} \log(2\pi e)^n |K| \quad \text{bits,}$$

where $|K|$ denotes the determinant of K . $f(\mathbf{x}) = \frac{1}{(\sqrt{2\pi})^n |K|^{\frac{1}{2}}} e^{-\frac{1}{2}(\mathbf{x}-\mu)^T K^{-1}(\mathbf{x}-\mu)}.$

Differential Entropy

$$[X_1 - \mu_1, X_2 - \mu_2, X_3 - \mu_3] \begin{bmatrix} K_{11} & K_{12} & K_{13} \\ K_{21} & K_{22} & K_{23} \\ K_{31} & K_{32} & K_{33} \end{bmatrix} \begin{bmatrix} X_1 - \mu_1 \\ X_2 - \mu_2 \\ X_3 - \mu_3 \end{bmatrix} e^{-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu})^T \mathbf{K}^{-1}(\mathbf{x} - \boldsymbol{\mu})}$$

$$\begin{aligned} h(f) &= - \int f(\mathbf{x}) \left[-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu})^T \mathbf{K}^{-1}(\mathbf{x} - \boldsymbol{\mu}) - \ln \left(\sqrt{2\pi} \right)^n |\mathbf{K}|^{\frac{1}{2}} \right] d\mathbf{x} \\ &= \frac{1}{2} E \left[\sum_{i,j} (X_i - \mu_i) (K^{-1})_{ij} (X_j - \mu_j) \right] + \frac{1}{2} \ln(2\pi)^n |\mathbf{K}| \\ &= \frac{1}{2} E \left[\sum_{i,j} (X_i - \mu_i)(X_j - \mu_j) (K^{-1})_{ij} \right] + \frac{1}{2} \ln(2\pi)^n |\mathbf{K}| \\ &= \frac{1}{2} \sum_{i,j} E[(X_j - \mu_j)(X_i - \mu_i)] (K^{-1})_{ij} + \frac{1}{2} \ln(2\pi)^n |\mathbf{K}| \\ &= \frac{1}{2} \sum_j \left[\sum_i K_{ji} (K^{-1})_{ij} \right] + \frac{1}{2} \ln(2\pi)^n |\mathbf{K}| \\ &= \frac{1}{2} \sum_j (K K^{-1})_{jj} + \frac{1}{2} \ln(2\pi)^n |\mathbf{K}| \end{aligned}$$

$$\begin{aligned} &= \frac{1}{2} \sum_j I_{jj} + \frac{1}{2} \ln(2\pi)^n |\mathbf{K}| \\ &= \frac{n}{2} + \frac{1}{2} \ln(2\pi)^n |\mathbf{K}| \\ &= \frac{1}{2} \ln(2\pi e)^n |\mathbf{K}| \quad \text{nats} \\ &= \frac{1}{2} \log(2\pi e)^n |\mathbf{K}| \quad \text{bits.} \end{aligned}$$

IIE, Xi'an Jiaotong University

9

9

Differential Entropy



□ Conditional Differential Entropy

Definition If X, Y have a joint density function $f(x, y)$, we can define the conditional differential entropy $h(X|Y)$ as

$$h(X|Y) = - \int f(x, y) \log f(x|y) dx dy.$$

Since in general $f(x|y) = f(x, y)/f(y)$, we can also write

$$h(X|Y) = h(X, Y) - h(Y).$$

Chain rule

IIE, Xi'an Jiaotong University

10

10



Mutual Information

Definition The mutual information $I(X; Y)$ between two random variables with joint density $f(x, y)$ is defined as

$$I(X; Y) = \int f(x, y) \log \frac{f(x, y)}{f(x)f(y)} dx dy.$$

From the definition it is clear that

$$I(X; Y) = h(X) - h(X|Y) = h(Y) - h(Y|X) = h(X) + h(Y) - h(X, Y)$$

□ Relationship (Venn diagram)

Theorem 2.4.1 (Mutual information and entropy)

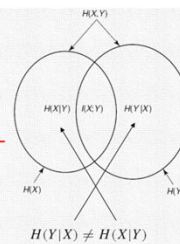
$$I(X; Y) = H(X) - H(X|Y)$$

$$I(X; Y) = H(Y) - H(Y|X)$$

$$I(X; Y) = H(X) + H(Y) - H(X, Y)$$

$$I(X; Y) = I(Y; X)$$

$$I(X; X) = H(X).$$



11

11



Mutual Information

Example 8.5.1 (Mutual information between correlated Gaussian random variables with correlation ρ) Let $(X, Y) \sim \mathcal{N}(0, K)$, where

$$K = \begin{bmatrix} \sigma^2 & \rho\sigma^2 \\ \rho\sigma^2 & \sigma^2 \end{bmatrix}.$$

Then $h(X) = h(Y) = \frac{1}{2} \log(2\pi e)\sigma^2$ and $h(X, Y) = \frac{1}{2} \log(2\pi e)^2 |K| = \frac{1}{2} \log(2\pi e)^2 \sigma^4 (1 - \rho^2)$, and therefore

$$I(X; Y) = h(X) + h(Y) - h(X, Y) = -\frac{1}{2} \log(1 - \rho^2).$$

If $\rho = 0$, X and Y are independent and the mutual information is 0. If $\rho = \pm 1$, X and Y are perfectly correlated and the mutual information is infinite.

12

12



Properties and Relations

□ Properties

$I(X; Y) \geq 0$ with equality iff X and Y are independent.

Proof:

$$\begin{aligned} -I(X; Y) &= \int f(x, y) \log \frac{f(x)f(y)}{f(x, y)} dx dy \\ &\leq \log \int f(x, y) \frac{f(x)f(y)}{f(x, y)} dx dy \\ &= \log 1 = 0 \end{aligned}$$

Corollary $h(X|Y) \leq h(X)$ with equality iff X and Y are independent.

微分熵可能是负，但是互信息不可能为负



Properties and Relations

■ Chain rule

Theorem 8.6.2 (Chain rule for differential entropy)

$$h(X_1, X_2, \dots, X_n) = \sum_{i=1}^n h(X_i | X_1, X_2, \dots, X_{i-1}).$$

■ Differential entropies of transformations

Theorem 8.6.3

$$h(X + c) = h(X).$$

Translation does not change the differential entropy.

Theorem 8.6.4

$$h(aX) = h(X) + \log |a|.$$



Properties and Relations

□ 坐标变换后概率密度函数的变化

$$\begin{cases} Y_1 = g_1(x_1, x_2, \dots, x_N) \\ Y_2 = g_2(x_1, x_2, \dots, x_N) \\ Y_N = g_N(x_1, x_2, \dots, x_N) \end{cases} \Leftrightarrow \begin{cases} X_1 = f_1(Y_1, Y_2, \dots, Y_N) \\ X_2 = f_2(Y_1, Y_2, \dots, Y_N) \\ X_N = f_N(Y_1, Y_2, \dots, Y_N) \end{cases}$$

$$p_{\mathbf{Y}}(y_1, y_2, \dots, y_N) = p_{\mathbf{X}}(x_1, x_2, \dots, x_N) \left| J \left(\frac{\mathbf{X}}{\mathbf{Y}} \right) \right|$$

$$J \left(\frac{\mathbf{X}}{\mathbf{Y}} \right) = \frac{\partial (X_1, X_2, \dots, X_N)}{\partial (Y_1, Y_2, \dots, Y_N)} = \begin{vmatrix} \frac{\partial f_1}{\partial Y_1} & \frac{\partial f_2}{\partial Y_1} & \frac{\partial f_N}{\partial Y_1} \\ \frac{\partial f_1}{\partial Y_2} & \frac{\partial f_2}{\partial Y_2} & \frac{\partial f_N}{\partial Y_2} \\ \frac{\partial f_1}{\partial Y_N} & \frac{\partial f_2}{\partial Y_N} & \frac{\partial f_N}{\partial Y_N} \end{vmatrix}$$



Properties and Relations

Proof: Let $Y = aX$. Then $f_Y(y) = \frac{1}{|a|} f_X\left(\frac{y}{a}\right)$, and

$$\begin{aligned} h(aX) &= - \int f_Y(y) \log f_Y(y) dy \\ &= - \int \frac{1}{|a|} f_X\left(\frac{y}{a}\right) \log \left(\frac{1}{|a|} f_X\left(\frac{y}{a}\right) \right) dy \\ &= - \int f_X(x) \log f_X(x) dx + \log |a| \\ &= h(X) + \log |a|, \end{aligned}$$

after a change of variables in the integral.

Corollary

$$h(A\mathbf{X}) = h(\mathbf{X}) + \log |\det(A)|.$$



Properties and Relations

- Multivariate normal distribution maximizes the entropy over all distributions with the same covariance.

Theorem 8.6.5 Let the random vector $\mathbf{X} \in \mathbf{R}^n$ have zero mean and covariance $K = E\mathbf{X}\mathbf{X}^T$ (i.e., $K_{ij} = EX_iX_j$, $1 \leq i, j \leq n$). Then $h(\mathbf{X}) \leq \frac{1}{2} \log(2\pi e)^n |K|$, with equality iff $\mathbf{X} \sim \mathcal{N}(0, K)$.

Proof: Let $g(\mathbf{x})$ be any density satisfying $\int g(\mathbf{x})x_ix_jd\mathbf{x} = K_{ij}$ for all i, j . Let ϕ_K be the density of a $\mathcal{N}(0, K)$ vector as given in (8.35), where we set $\mu = 0$. Note that $\log \phi_K(\mathbf{x})$ is a quadratic form and $\int x_ix_j\phi_K(\mathbf{x})d\mathbf{x} = K_{ij}$. Then

$$\begin{aligned} \int_S f \log \frac{g}{f} & 0 \leq \int g \log(g/\phi_K) & \text{Jensen's inequality} \\ \leq \log \int_S f \frac{g}{f} & = -h(g) - \int g \log \phi_K & \phi_K(\mathbf{x}) = \frac{1}{(\sqrt{2\pi})^n |K|^{\frac{1}{2}}} e^{-\frac{1}{2}(\mathbf{x}-\mu)^T K^{-1}(\mathbf{x}-\mu)} \\ = \log \int_S g & \\ \leq \log 1 = 0. & \end{aligned}$$



Properties and Relations

$$\begin{aligned} & = -h(g) - \int g \log \phi_K \\ & = -h(g) - \int \phi_K \log \phi_K \\ & = -h(g) + h(\phi_K), \end{aligned}$$

where the substitution $\int g \log \phi_K = \int \phi_K \log \phi_K$ follows from the fact that g and ϕ_K yield the same moments of the quadratic form $\log \phi_K(\mathbf{x})$. □

$$\log \phi_K(\mathbf{x}) = \left[-\frac{1}{2}(\mathbf{x} - \mu)^T K^{-1}(\mathbf{x} - \mu) - \ln \left((\sqrt{2\pi})^n |K|^{\frac{1}{2}} \right) \right]$$



Properties and Relations

□ Relation of differential entropy and discrete entropy

- Consider a random variable X with density $f(x)$.
- Divide the range of X into bins of length Δ . There exists a value x_i within each bin such that

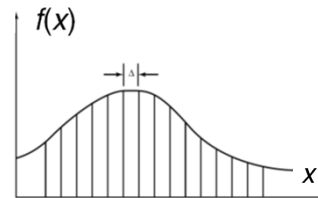
$$f(x_i)\Delta = \int_{i\Delta}^{(i+1)\Delta} f(x) dx.$$

- Quantize random variable

$$X^\Delta = x_i \quad \text{if } i\Delta \leq X < (i+1)\Delta$$

Then the probability that $X^\Delta = x_i$

$$p_i = \int_{i\Delta}^{(i+1)\Delta} f(x) dx = f(x_i)\Delta.$$



Now let us calculate the entropy of the quantized version



Properties and Relations

□ Quantized entropy

$$H(X^\Delta) = - \sum_{-\infty}^{\infty} p_i \log p_i$$

$$= - \sum_{-\infty}^{\infty} f(x_i)\Delta \log(f(x_i)\Delta)$$

$$\sum f(x_i)\Delta = \int f(x) = 1$$

$$\begin{aligned} \int -f(x) \log f(x) &= - \sum \Delta f(x_i) \log f(x_i) - \sum f(x_i)\Delta \log \Delta \\ \Delta \rightarrow 0 &\quad \quad \quad = - \sum \Delta f(x_i) \log f(x_i) - \log \Delta. \end{aligned}$$

Theorem 8.3.1 If the density $f(x)$ of the random variable X is Riemann integrable, then

$$H(X^\Delta) + \log \Delta \rightarrow h(f) = h(X), \quad \text{as } \Delta \rightarrow 0.$$

Thus, the entropy of an n -bit quantization of a continuous random variable X is approximately $h(X) + n$.



Properties and Relations

□ Some examples

1. If X has a uniform distribution on $[0, 1]$ and we let $\Delta = 2^{-n}$, then $h = 0$, $H(X^\Delta) = n$, and n bits suffice to describe X to n bit accuracy.
2. If X is uniformly distributed on $[0, \frac{1}{8}]$, the first 3 bits to the right of the decimal point must be 0. To describe X to n -bit accuracy requires only $n - 3$ bits, which agrees with $h(X) = -3$.
3. If $X \sim \mathcal{N}(0, \sigma^2)$ with $\sigma^2 = 100$, describing X to n bit accuracy would require on the average $n + \frac{1}{2} \log(2\pi e \sigma^2) = n + 5.37$ bits.

In general, $h(X) + n$ is the number of bits *on the average* required to describe X to n -bit accuracy.



AEP

□ AEP for a continuous random variable

Theorem 8.2.1 Let $X_1, X_2, \dots, X_n$ be a sequence of random variables drawn i.i.d. according to the density $f(x)$. Then

$$-\frac{1}{n} \log f(X_1, X_2, \dots, X_n) \rightarrow E[-\log f(X)] = h(X) \quad \text{in probability.}$$

Weak law of large numbers.

□ Typical set

Definition For $\epsilon > 0$ and any n , we define the *typical set* $A_\epsilon^{(n)}$ with respect to $f(x)$ as follows:

$$A_\epsilon^{(n)} = \left\{ (x_1, x_2, \dots, x_n) \in S^n : \left| -\frac{1}{n} \log f(x_1, x_2, \dots, x_n) - h(X) \right| \leq \epsilon \right\},$$

where $f(x_1, x_2, \dots, x_n) = \prod_{i=1}^n f(x_i)$.



AEP

□ Volume

- The volume of the typical set for continuous random variables is the analog of the cardinality of the typical set for the discrete case

Definition The volume $\text{Vol}(A)$ of a set $A \subset \mathcal{R}^n$ is defined as

$$\text{Vol}(A) = \int_A dx_1 dx_2 \cdots dx_n.$$

□ Properties of AEP

The properties of the typical set for continuous random variables parallel those for discrete random variables.

Theorem 8.2.2 The typical set $A_\epsilon^{(n)}$ has the following properties:

1. $\Pr(A_\epsilon^{(n)}) > 1 - \epsilon$ for n sufficiently large.
2. $\text{Vol}(A_\epsilon^{(n)}) \leq 2^{n(h(X)+\epsilon)}$ for all n .
3. $\text{Vol}(A_\epsilon^{(n)}) \geq (1 - \epsilon)2^{n(h(X)-\epsilon)}$ for n sufficiently large.



AEP

- By AEP for a continuous random variable, immediately we have property 1.

- Also,

$$\begin{aligned} 1 &= \int_{S^n} f(x_1, x_2, \dots, x_n) dx_1 dx_2 \cdots dx_n \\ &\geq \int_{A_\epsilon^{(n)}} f(x_1, x_2, \dots, x_n) dx_1 dx_2 \cdots dx_n \\ &\geq \int_{A_\epsilon^{(n)}} 2^{-n(h(X)+\epsilon)} dx_1 dx_2 \cdots dx_n \\ &= 2^{-n(h(X)+\epsilon)} \int_{A_\epsilon^{(n)}} dx_1 dx_2 \cdots dx_n \\ &= 2^{-n(h(X)+\epsilon)} \text{Vol}(A_\epsilon^{(n)}). \end{aligned}$$

Hence we have property 2. the volume of the typical set is at least this large.



AEP

- If n is sufficiently large so that property 1 is satisfied, then

$$\begin{aligned} 1 - \epsilon &\leq \int_{A_\epsilon^{(n)}} f(x_1, x_2, \dots, x_n) dx_1 dx_2 \cdots dx_n \\ &\leq \int_{A_\epsilon^{(n)}} 2^{-n(h(X) - \epsilon)} dx_1 dx_2 \cdots dx_n \\ &= 2^{-n(h(X) - \epsilon)} \int_{A_\epsilon^{(n)}} dx_1 dx_2 \cdots dx_n \\ &= 2^{-n(h(X) - \epsilon)} \text{Vol}(A_\epsilon^{(n)}), \end{aligned}$$

establishing property 3.

$$(1 - \epsilon) 2^{n(h(X) - \epsilon)} \leq \text{Vol}(A_\epsilon^{(n)}) \leq 2^{n(h(X) + \epsilon)}.$$

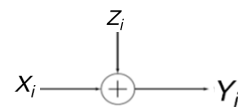


Gaussian Channel

- Additive Gaussian white noise channel

$$Y_i = X_i + Z_i, \quad Z_i \sim \mathcal{N}(0, N).$$

independent



- noise variance = 0, the receiver receives the transmitted symbol perfectly $\rightarrow$ infinite capacity.
- noise variance is nonzero, but there is no constraint on the input. Using infinite power $\rightarrow$ infinite capacity.
- There is an average energy or power constraint on the input

$$\frac{1}{n} \sum_{i=1}^n x_i^2 \leq P.$$

Then, what is the capacity ??



Channel Capacity of Gaussian Channel

Definition

- Maximum of the mutual information between the input and output over all distributions on the input that satisfy the power constraint.

Definition The *information capacity* of the Gaussian channel with power constraint P is

$$C = \max_{f(x): E X^2 \leq P} I(X; Y).$$

Derivation

$$\begin{aligned}
 I(X; Y) &= h(Y) - h(Y|X) \\
 &= h(Y) - h(X + Z|X) \\
 &= h(Y) - h(Z|X) \quad \leftarrow \text{Z is independent of X.} \\
 &= h(Y) - h(Z), \quad \leftarrow h(Z) = \frac{1}{2} \log 2\pi e N. \\
 EY^2 &= E(X + Z)^2 \\
 &= EX^2 + 2EXEZ + EZ^2 \\
 &= P + N, \quad \leftarrow \leq \frac{1}{2} \log 2\pi e (P + N)
 \end{aligned}$$



Channel Capacity of Gaussian Channel

$$\begin{aligned}
 I(X; Y) &\leq \frac{1}{2} \log 2\pi e (P + N) - \frac{1}{2} \log 2\pi e N \\
 &= \frac{1}{2} \log \left(1 + \frac{P}{N} \right).
 \end{aligned}$$

$$C = \max_{EX^2 \leq P} I(X; Y) = \frac{1}{2} \log \left(1 + \frac{P}{N} \right),$$

and the maximum is attained when $X \sim \mathcal{N}(0, P)$.

The above derivation is for **information channel capacity**.

We will now show that this capacity is also the **operational channel capacity**
 → the supremum of the rates achievable for the channel.



Channel Capacity of Gaussian Channel

□ Code definition

Definition An (M, n) code for the Gaussian channel with power constraint P consists of the following:

1. An index set $\{1, 2, \dots, M\}$.
2. An encoding function $x : \{1, 2, \dots, M\} \rightarrow \mathcal{X}^n$, yielding codewords $x^n(1), x^n(2), \dots, x^n(M)$, satisfying the power constraint P ; that is, for every codeword

$$\sum_{i=1}^n x_i^2(w) \leq nP, \quad w = 1, 2, \dots, M.$$

3. A decoding function

$$g : \mathcal{Y}^n \rightarrow \{1, 2, \dots, M\}.$$

The arithmetic average of the probability of error is $P_e^{(n)} = \frac{1}{2^{nR}} \sum \lambda_i.$



Channel Capacity of Gaussian Channel

□ Channel Capacity of Gaussian Channel

Definition A rate R is said to be *achievable* for a Gaussian channel with a power constraint P if there exists a sequence of $(2^{nR}, n)$ codes with codewords satisfying the power constraint such that the maximal probability of error $\lambda^{(n)}$ tends to zero. The capacity of the channel is the supremum of the achievable rates.

(0, 1) [0, 1]

Theorem 9.1.1 The capacity of a Gaussian channel with power constraint P and noise variance N is

$$C = \frac{1}{2} \log \left(1 + \frac{P}{N} \right) \quad \text{bits per transmission.}$$

We first present the intuition why we may be able to construct a code with a low probability of error.



Channel Capacity of Gaussian Channel

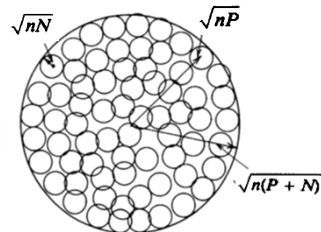
□ Sphere packing interpretation

Consider any codeword of length n . The received vector is normally distributed with mean equal to the true codeword and variance equal to the noise variance. With high probability, the received vector is contained in a sphere of radius $\sqrt{n(N + \epsilon)}$ around the true codeword.

How many such codewords can we choose?

The volume of an n -dimensional sphere $C_n r^n$, r is the radius.

The received vectors have energy no greater than $n(P + N)$, so they lie in a sphere of radius $\sqrt{n(P + N)}$.



$$\frac{C_n(n(P + N))^{\frac{n}{2}}}{C_n(nN)^{\frac{n}{2}}} = 2^{\frac{n}{2} \log \left(1 + \frac{P}{N}\right)}$$



Channel Capacity of Gaussian Channel

□ Achievability Proof

- the same ideas as in the proof of the channel coding theorem in the case of discrete channels is used.
- random codes and joint typicality decoding
- Difference: 1) power constraint and 2) the variables are continuous.

1. *Generation of the codebook.* We wish to generate a codebook in which all the codewords satisfy the power constraint. To ensure this, we generate the codewords with each element i.i.d. according to a normal distribution with variance $P - \epsilon$. Since for large n , $\frac{1}{n} \sum X_i^2 \rightarrow P - \epsilon$, the probability that a codeword does not satisfy the power constraint will be small. Let $X_i(w)$, $i = 1, 2, \dots, n$, $w = 1, 2, \dots, 2^{nR}$ be i.i.d. $\sim \mathcal{N}(0, P - \epsilon)$, forming codewords $X^n(1), X^n(2), \dots, X^n(2^{nR}) \in \mathcal{R}^n$.



Channel Capacity of Gaussian Channel

2. *Encoding*. After the generation of the codebook, the codebook is revealed to both the sender and the receiver. To send the message index w , the transmitter sends the w th codeword $X^n(w)$ in the codebook.
3. *Decoding*. The receiver looks down the list of codewords $\{X^n(w)\}$ and searches for one that is jointly typical with the received vector. If there is one and only one such codeword $X^n(w)$, the receiver declares $\hat{W} = w$ to be the transmitted codeword. Otherwise, the receiver declares an error. The receiver also declares an error if the chosen codeword does not satisfy the power constraint.
4. *Probability of error*. Without loss of generality, assume that codeword 1 was sent. Thus, $Y^n = X^n(1) + Z^n$. Define the following events:

$$E_0 = \left\{ \frac{1}{n} \sum_{j=1}^n X_j^2(1) > P \right\} \quad \text{and} \quad E_i = \{(X^n(i), Y^n) \text{ is in } A_\epsilon^{(n)}\}.$$



Channel Capacity of Gaussian Channel

□ Then an error occurs if

- the power constraint is violated E_0
- the transmitted codeword and the received sequence are not jointly typical E_1^c
- some wrong codeword is jointly typical with the received sequence

$$E_2 \cup E_3 \cup \dots \cup E_{2^{nR}}$$

$$\Pr(\mathcal{E} | W = 1) = P(\mathcal{E}) = P(E_0 \cup E_1^c \cup E_2 \cup E_3 \cup \dots \cup E_{2^{nR}})$$

$$\leq P(E_0) + P(E_1^c) + \sum_{i=2}^{2^{nR}} P(E_i),$$

$$P(E_0) \rightarrow 0 \text{ as } n \rightarrow \infty. \quad \text{by the law of large numbers}$$

$$P(E_1^c) \leq \epsilon \quad \text{by the joint AEP}$$



Channel Capacity of Gaussian Channel

Since by the code generation process, $X^n(1)$ and $X^n(i)$ are independent, so are Y^n and $X^n(i)$. Hence, the probability that $X^n(i)$ and Y^n will be jointly typical is $\leq 2^{-n(I(X;Y)-3\epsilon)}$ by the joint AEP.

$$\begin{aligned}
 \Pr(\mathcal{E}) &= \frac{1}{2^{nR}} \sum \lambda_i = P_e^{(n)} = \Pr(\mathcal{E}) = \Pr(\mathcal{E}|W=1) \\
 &\leq P(E_0) + P(E_1^c) + \sum_{i=2}^{2^{nR}} P(E_i) \\
 &\leq \epsilon + \epsilon + \sum_{i=2}^{2^{nR}} 2^{-n(I(X;Y)-3\epsilon)} \\
 &= 2\epsilon + (2^{nR} - 1) 2^{-n(I(X;Y)-3\epsilon)} \quad \text{for } n \text{ sufficiently large} \\
 &\leq 2\epsilon + 2^{3n\epsilon} 2^{-n(I(X;Y)-R)} \leq 3\epsilon \quad \text{and } R < I(X;Y) - 3\epsilon.
 \end{aligned}$$



Channel Capacity of Gaussian Channel

□ Converse $\rightarrow$ rates $R > C$ are not achievable.

Proof: (Converse to Theorem 9.1.1). We must show that if $P_e^{(n)} \rightarrow 0$ for a sequence of $(2^{nR}, n)$ codes for a Gaussian channel with power constraint P , then

$$R \leq C = \frac{1}{2} \log \left(1 + \frac{P}{N} \right).$$

Consider any $(2^{nR}, n)$ code that satisfies the power constraint, that is,

$$\frac{1}{n} \sum_{i=1}^n x_i^2(w) \leq P,$$

for $w = 1, 2, \dots, 2^{nR}$.

Proceeding as in the converse for the discrete case



Channel Capacity of Gaussian Channel

- Apply Fano's inequality

$$W \rightarrow X^n(W) \rightarrow Y^n \rightarrow \hat{W}.$$

$$H(W|\hat{W}) \leq 1 + nRP_e^{(n)} = n\epsilon_n,$$

$$nR = H(W) = I(W; \hat{W}) + H(W|\hat{W})$$

$$\leq I(W; \hat{W}) + n\epsilon_n \quad \rightarrow \text{Fano's inequality}$$

$$\leq I(X^n; Y^n) + n\epsilon_n \quad \rightarrow \text{the data-processing inequality}$$

$$= h(Y^n) - h(Y^n|X^n) + n\epsilon_n$$

$$= h(Y^n) - h(Z^n) + n\epsilon_n$$

$$\leq \sum_{i=1}^n h(Y_i) - h(Z^n) + n\epsilon_n$$

$$= \sum_{i=1}^n h(Y_i) - \sum_{i=1}^n h(Z_i) + n\epsilon_n = \sum_{i=1}^n I(X_i; Y_i) + n\epsilon_n.$$



Channel Capacity of Gaussian Channel

- Now let P_i be the average power of the i th column of the codebook, that is,

$$P_i = \frac{1}{2nR} \sum_w x_i^2(w).$$

Then, since $Y_i = X_i + Z_i$ and since X_i and Z_i are independent, the average power EY_i^2 of Y_i is $P_i + N$. Hence, since entropy is maximized by the normal distribution,

$$h(Y_i) \leq \frac{1}{2} \log 2\pi e(P_i + N).$$

$$nR \leq \sum (h(Y_i) - h(Z_i)) + n\epsilon_n$$

$$\leq \sum \left(\frac{1}{2} \log(2\pi e(P_i + N)) - \frac{1}{2} \log 2\pi eN \right) + n\epsilon_n$$

$$= \sum \frac{1}{2} \log \left(1 + \frac{P_i}{N} \right) + n\epsilon_n.$$



Channel Capacity of Gaussian Channel

- Since each of the codewords satisfies the power constraint, so does their average

$$\frac{1}{n} \sum_i P_i \leq P.$$

- Since $f(x) = \frac{1}{2} \log(1+x)$ is a concave function of x , from the definition of the concave function

$$\begin{aligned} \frac{1}{n} \sum_{i=1}^n \frac{1}{2} \log \left(1 + \frac{P_i}{N} \right) &\leq \frac{1}{2} \log \left(1 + \frac{1}{n} \sum_{i=1}^n \frac{P_i}{N} \right) \\ &\leq \frac{1}{2} \log \left(1 + \frac{P}{N} \right). \end{aligned}$$

$$R \leq \frac{1}{2} \log \left(1 + \frac{P}{N} \right) + \epsilon_n, \epsilon_n \rightarrow 0, \text{ and we have the required converse.}$$



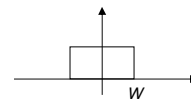
Some other Channels

- Band-limited Channel

- Band-limited channel with white noise

$$Y(t) = (X(t) + Z(t)) * h(t),$$

convolution



- $h(t)$ is the impulse response of an ideal bandpass filter, which cuts out all frequencies greater than W . (telephone line)
- Nyquist Theorem
- Sampling a bandlimited signal at a sampling rate $1/2W$ is sufficient to reconstruct the signal from the samples.

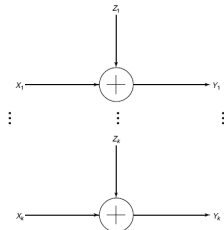
time interval $[0, T]$ Energy per sample $PT/2WT$ the capacity
per sample $2W$ samples
each secondnoise variance per sample $\frac{N_0}{2} 2W \frac{T}{2WT}$

$$C = \frac{1}{2} \log \left(1 + \frac{P}{N} \right) \rightarrow \frac{1}{2} \log \left(1 + \frac{P}{N_0 W} \right) \rightarrow C = W \log \left(1 + \frac{P}{N_0 W} \right)$$



Some other Channels

□ Parallel Gaussian Channel



$$Y_j = X_j + Z_j, \quad j = 1, 2, \dots, k,$$

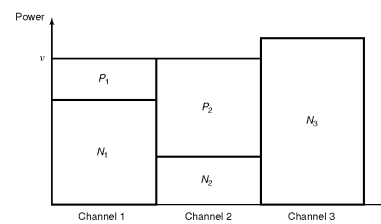
$$Z_j \sim \mathcal{N}(0, N_j),$$

$$E \sum_{j=1}^k X_j^2 \leq P.$$

$$C = \max_{f(x_1, x_2, \dots, x_k): \sum E X_j^2 \leq P} I(X_1, X_2, \dots, X_k; Y_1, Y_2, \dots, Y_k).$$

The result shows that better channel, more power.-- water-filling

In this course, we will not investigate more into this problem



~The end~